

Are oral contraceptives a significant contributor to the estrogenicity of drinking water?

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Recent observed feminization of aquatic animals has raised concerns about estrogenic compounds in water supplies and the potential for these chemicals to reach drinking water. Public perception frequently attributes this feminization to oral contraceptives (OCs) in wastewater and raises concerns that exposure to OCs in drinking water may contribute to the recent rise in human reproductive problems. This paper reviews the literature regarding various sources of estrogens, in surface, source and drinking water, with an emphasis on the active molecule that comes from OCs. It includes discussion of the various agricultural, industrial, and municipal sources and outlines the contributions of estrogenic chemicals to the estrogenicity of waterways and estimates that the risk of exposure to synthetic estrogens in drinking water on human health is negligible. This paper also provides recommendations for strategies to better understand all the potential sources of estrogenic compounds in the environment and possibilities to reduce the levels of estrogenic chemicals in the water supply.